

MENG 3020 – Midterm Exam Solution – Fall 2025

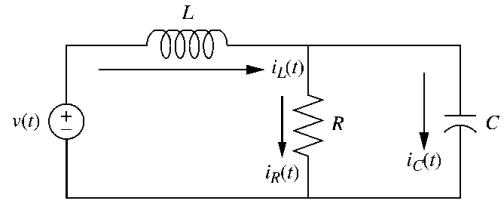
Question 1. [15 Marks] *Modeling of electrical systems.*

Consider the following RLC network with the following parameters:

$$R = 6 \, \Omega$$

$$L = 2 \, H$$

$$C = 0.25 \, F$$



a) [5 marks] Redraw the RLC network in Laplace domain by applying the **complex impedance** method. Show the impedances and the variables in Laplace domain on the circuit.

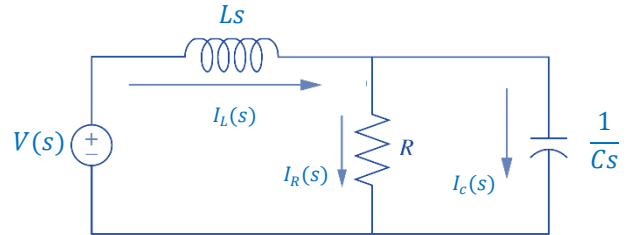
Replace the passive element values with their impedances.

Replace all sources and variables with their Laplace transform.

Impedance of inductance $\rightarrow Ls = 2s$

Impedance of capacitor $\rightarrow \frac{1}{Cs} = \frac{1}{0.25s} = \frac{4}{s}$

Impedance of resistances same of their resistance values.



b) [10 marks] Apply KVL and KCL or any other circuit analysis method to find the transfer function $I_L(s)/V(s)$. Show your work.

Method 1:

Apply KVL in left loop:

$$\text{KVL} \rightarrow V(s) = V_L(s) + V_R(s) \rightarrow V(s) = LsI_L(s) + RI_R(s) \quad \text{Eqn. (1)}$$

Apply KCL at the top node:

$$\text{KCL} \rightarrow I_L(s) = I_R(s) + I_C(s) \rightarrow I_R(s) = I_L(s) - I_C(s) = I_L(s) - \frac{V_C(s)}{\frac{1}{Cs}} = I_L(s) - CsV_C(s) \quad \text{Eqn. (2)}$$

Since the resistor and the capacitor are in parallel: $V_C(s) = V_R(s)$

Substituting $V_C(s) = V_R(s)$ in Eqn. (2) we have

$$I_R(s) = I_L(s) - CsV_R(s) = I_L(s) - CsRI_R(s) \rightarrow (1 + RCs)I_R(s) = I_L(s) \rightarrow I_R(s) = \frac{1}{RCs + 1}I_L(s)$$

Substitute $I_R(s)$ in Eqn. (1)

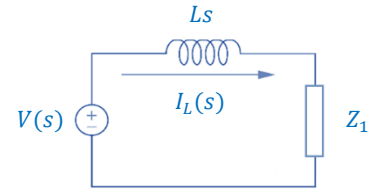
$$V(s) = LsI_L(s) + \frac{R}{RCs + 1}I_L(s) = \left(Ls + \frac{R}{RCs + 1} \right) I_L(s) = \left(\frac{RLCs^2 + Ls + R}{RCs + 1} \right) I_L(s)$$

The transfer function is:
$$\rightarrow \frac{I_L(s)}{V(s)} = \frac{RCs + 1}{RLCs^2 + Ls + R} = \frac{1.5s + 1}{3s^2 + 2s + 6}$$

Method 2:

Find the total impedance Z_1 of the parallel resistor and capacitor:

$$Z_1 = \frac{\frac{R}{Cs}}{R + \frac{1}{Cs}} = \frac{R}{RCs + 1}$$

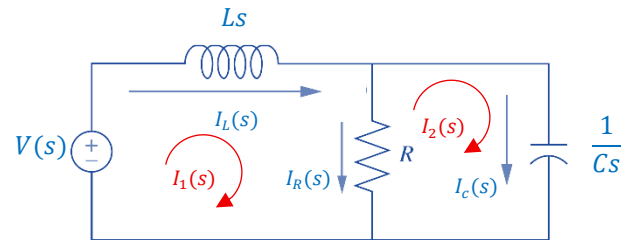


Apply KVL in the loop:

$$\text{KVL} \rightarrow V(s) = V_L(s) + V_{Z_1}(s) \rightarrow V(s) = LsI_L(s) + Z_1I_L(s) \rightarrow V(s) = (Ls + Z_1)I_L(s) \quad \text{Eqn. (3)}$$

Substitute Z_1 in Eqn. (3) and form the transfer function:

$$\text{The transfer function is:} \rightarrow \frac{I_L(s)}{V(s)} = \frac{1}{Ls + Z_1} = \frac{1}{Ls + \frac{R}{RCs+1}} = \frac{RCs+1}{RLCs^2 + Ls + R} = \frac{1.5s+1}{3s^2+2s+6}$$

Method 3:

Mesh Analysis:

$$\text{Mesh 1} \rightarrow LsI_1(s) + R(I_1(s) - I_2(s)) - V(s) = 0 \rightarrow (Ls + R)I_1(s) - RI_2(s) = V(s) \quad \text{Eqn. (1)}$$

$$\text{Mesh 2} \rightarrow \frac{1}{Cs}I_2(s) + R(I_2(s) - I_1(s)) = 0 \rightarrow \left(\frac{1}{Cs} + R\right)I_2(s) - RI_1(s) = 0 \rightarrow \left(\frac{1+RCs}{Cs}\right)I_2(s) - RI_1(s) = 0 \quad \text{Eqn. (2)}$$

Since: $I_1(s) = I_L(s)$, find $I_2(s)$ from Eqn. (2) and substitute in Eqn. (1).

$$\text{From Eqn. (2): } I_2(s) = \left(\frac{RCs}{1+RCs}\right)I_1(s)$$

Substitute $I_2(s)$ in Eqn. (1)

$$(Ls + R)I_1(s) - R\left(\frac{RCs}{1+RCs}\right)I_1(s) = V(s) \rightarrow \left(\frac{(Ls + R)(1 + RCs) - R^2Cs}{1 + RCs}\right)I_1(s) = V(s)$$

$$\rightarrow \left(\frac{Ls + RLCs + R + R^2Cs - R^2Cs}{1 + RCs}\right)I_1(s) = V(s) \rightarrow \left(\frac{Ls + RLCs + R}{1 + RCs}\right)I_1(s) = V(s)$$

$$\text{The transfer function is:} \rightarrow \frac{I_L(s)}{V(s)} = \frac{RCs+1}{RLCs^2 + Ls + R} = \frac{1.5s+1}{3s^2+2s+6}$$

Question 2. [10 Marks] Transfer function properties and state-space model realization.

Consider the following transfer function model of a dynamic system,

$$\frac{Y(s)}{U(s)} = \frac{8s + 2}{s^2 + 11s + 5}$$

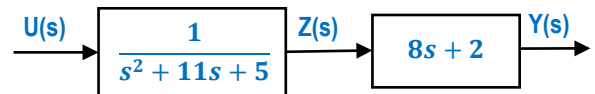
a) [2 marks] Determine the system order and clarify if the transfer function is proper or strictly proper. Justify your answer.

System order is the highest order of the denominator polynomial. Thus the system is **second order**.

Since order of the numerator is less than the order of the denominator, the system is **strictly proper**.

b) [8 marks] Obtain a state-space representation of the system by selecting the state variables as the phase variables. Find the state equations and the output equation. Show the results in matrix-vector form. Show all your steps.

$$\frac{Y(s)}{U(s)} = \frac{Z(s)}{U(s)} \times \frac{Y(s)}{Z(s)} = \frac{1}{s^2 + 11s + 5} \times (8s + 2)$$



First, find the state equations from the part with denominator.

$$s^2 Z(s) + 11sZ(s) + 5Z(s) = U(s) \quad \rightarrow \quad \ddot{z}(t) + 11\dot{z}(t) + 5z(t) = u(t)$$

Define the state variables as the phase variables:

$$q_1(t) = z(t) \quad \rightarrow \quad \dot{q}_1(t) = \dot{z}(t) \quad \rightarrow \quad \dot{q}_1(t) = q_2(t)$$

$$q_2(t) = \dot{z}(t) \quad \rightarrow \quad \dot{q}_2(t) = \ddot{z}(t) \quad \rightarrow \quad \dot{q}_2(t) = -11\dot{z}(t) - 5z(t) + u(t) \quad \rightarrow \quad \dot{q}_2(t) = -11q_2(t) - 5q_1(t) + u(t)$$

Find the output equation by considering the effect of the numerator term:

$$Y(s) = (8s + 2)Z(s) \quad \rightarrow \quad Y(s) = 8sZ(s) + 2Z(s) \quad \rightarrow \quad y(t) = 8\dot{z}(t) + 2z(t) \quad \rightarrow \quad y(t) = 8q_2(t) + 2q_1(t)$$

Therefore, the state equations and the output equation are obtained as follows:

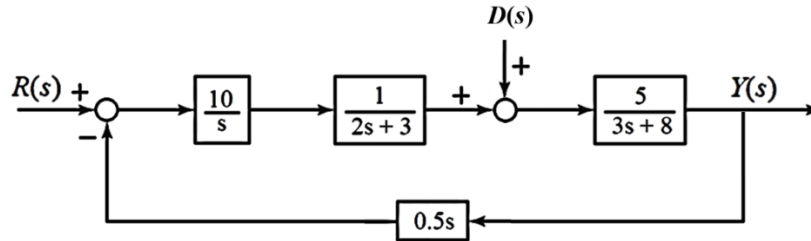
$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t) \quad \rightarrow \quad \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & -11 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}u(t) \quad \rightarrow \quad y(t) = \begin{bmatrix} 2 & 8 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + [0]u(t)$$

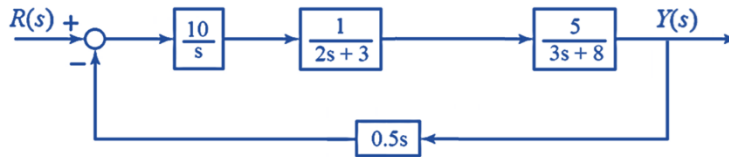
Question 3. [10 Marks] Block diagram model simplification.

Applying **Superposition Principles**, simplify the following block diagram and find the following transfer functions $G_1(s)$ and $G_2(s)$ in the simplest standard form. Show your work and the steps.

$$G_1(s) = \frac{Y(s)}{R(s)}, \quad G_2(s) = \frac{Y(s)}{D(s)}$$



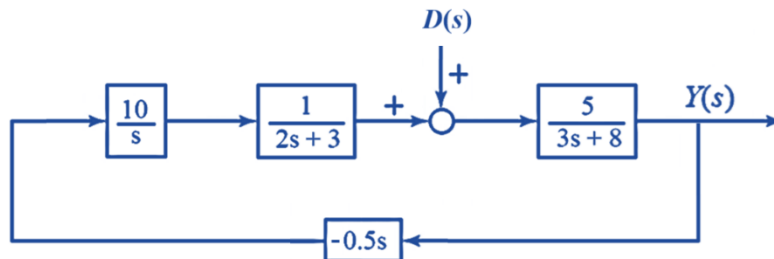
Applying superposition principles, set $D(s) = 0$ and find transfer function $G_1(s)$



$$G_1(s) = \frac{Y(s)}{R(s)} = \frac{\left(\frac{10}{s}\right)\left(\frac{1}{2s+3}\right)\left(\frac{5}{3s+8}\right)}{1 + \left(\frac{10}{s}\right)\left(\frac{1}{2s+3}\right)\left(\frac{5}{3s+8}\right)(0.5s)} = \frac{\frac{50}{s(2s+3)(3s+8)}}{1 + \frac{25s}{s(2s+3)(3s+8)}} = \frac{50}{s(2s+3)(3s+8) + 25s}$$

$$G_1(s) = \frac{50}{s(6s^2 + 25s + 49)}$$

Applying superposition principles, set $R(s) = 0$ and find transfer function $G_2(s)$



$$G_2(s) = \frac{Y(s)}{D(s)} = \frac{\frac{5}{3s+8}}{1 - \left(\frac{5}{3s+8}\right)\left(\frac{10}{s}\right)\left(\frac{1}{2s+3}\right)(-0.5s)} = \frac{\frac{5}{3s+8}}{1 + \frac{25s}{s(2s+3)(3s+8)}} = \frac{5s(2s+3)}{s(2s+3)(3s+8) + 25s}$$

$$G_2(s) = \frac{5s(2s+3)}{s(6s^2 + 25s + 49)} = \frac{5(2s+3)}{(6s^2 + 25s + 49)}$$

Question 4. [15 marks] Modeling of rotational mechanical systems.

Consider the following rotational mechanical system with the following parameters:

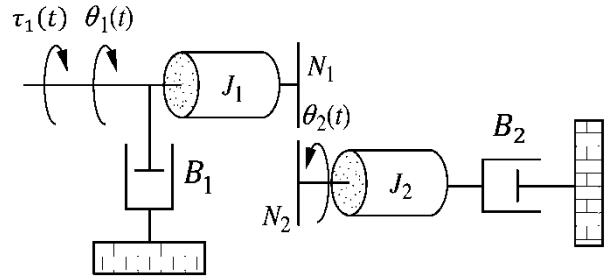
$$\frac{N_2}{N_1} = 10$$

$$J_1 = 5 \text{ kg.m}^2$$

$$B_1 = 2 \text{ N.m.s/rad}$$

$$J_2 = 700 \text{ kg.m}^2$$

$$B_2 = 800 \text{ N.m.s/rad}$$



a) [2 marks] Find the equivalent inertia J_e and the equivalent damping coefficient B_e in the output-side (side 2). Show your work.

The equivalent inertia in side 2 is:

$$J_e = J_1 \left(\frac{N_2}{N_1} \right)^2 + J_2 = 100J_1 + J_2 = 100(5) + 700 = 1200 \text{ kg.m}^2$$

The equivalent damping coefficient in side 2 is:

$$B_e = B_1 \left(\frac{N_2}{N_1} \right)^2 + B_2 = 100B_1 + B_2 = 100(2) + 800 = 1000 \text{ kg.m}^2$$

b) [2 marks] Find the reflected torque τ_1 , and reflected angular displacement θ_1 , from the input-side (side 1) to the output-side (side 2) by considering the effect of the gear ratio. Show your work.

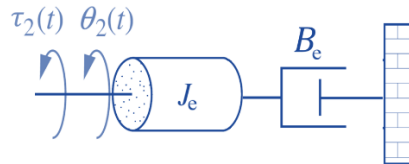
The reflected torque from side 1 to side 2 is:

$$\tau_2 = \left(\frac{N_2}{N_1} \right) \tau_1 = 10\tau_1$$

The reflected angular displacement from side 1 to side 2 is:

$$\theta_2 = \left(\frac{N_1}{N_2} \right) \theta_1 = 0.1\theta_1$$

c) [4 marks] Redraw the equivalent gearless system based on the reflected values. Show the reflected values on the system.



d) [7 marks] Derive the equation of motion and find the transfer function, $\theta_2(s)/T_1(s)$ for the gearless system. Show your work.

The equation of motion is:

$$\tau_2(t) - B_e \dot{\theta}_2(t) = J_e \ddot{\theta}_2(t) \rightarrow N\tau_1(t) - B_e \dot{\theta}_2(t) = J_e \ddot{\theta}_2(t)$$

Take Laplace transform with zero initial conditions to find the transfer function:

$$NT_1(s) = B_e s \theta_2(s) + J_e s^2 \theta_2(s) = (B_e s + J_e s^2) \theta_2(s)$$

$$\frac{\theta_2(s)}{T_1(s)} = \frac{N}{J_e s^2 + B_e s} = \frac{10}{s(1200s + 1000)}$$